

Equivalent Sources Interpolation Method

Now that we have corrected the gravity values for all known physical factors (Latitude, Free-air, and Terrain/Bouguer effects), we have obtained the Full **Topography-free Disturbance**.

However, these values are still spatially 'stuck' to the irregular ground surface where they were measured. To create a scientifically sound map, we now need to **harmonize the data spatially**.

Because gravity weakens and spreads out over distance, a coastal station at 50m records a sharp, intense signal from underground geology, while a station at the 3700m Teide summit measures a much smoother, weaker version of that exact same geology. To fix this spatial mismatch, the equivalent sources algorithm calculates a layer of virtual underground masses whose combined gravitational pull perfectly recreates your original observed values at their true, varying elevations. Once those virtual masses are mathematically established, the model uses Newton's laws to recalculate and project their gravity field onto a single, flat elevation plane, making every data point directly comparable.

Multiple methods exist, but the one we used is the classic equivalent source method where 1 source is placed below each datapoint. [Equivalent Sources | Harmonica v0.7.0](#)

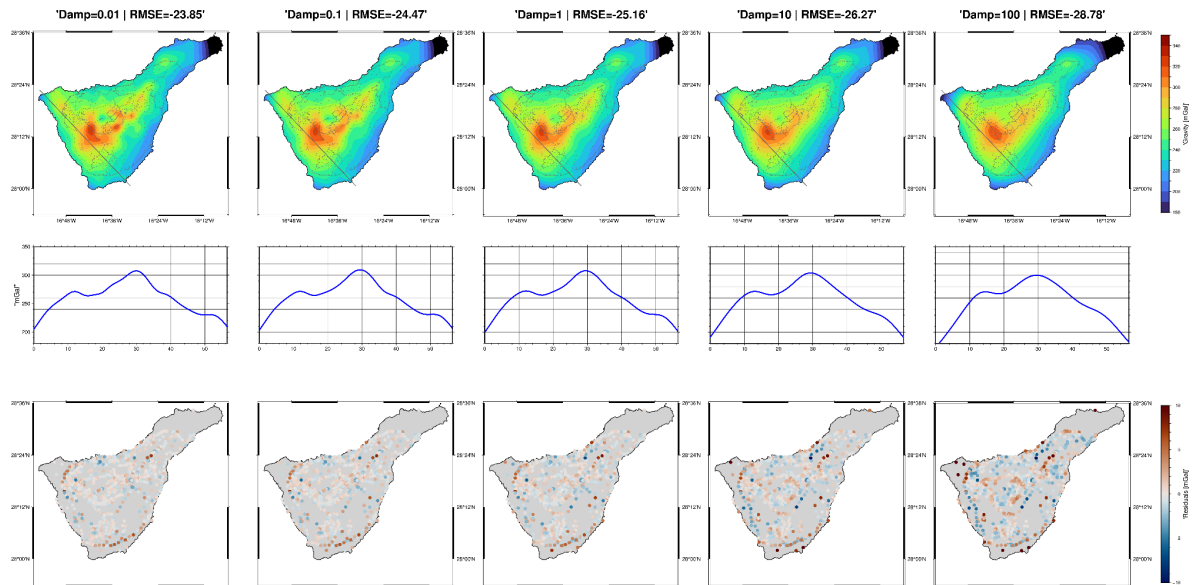
The hyperparameters needed to be chosen for the process.

They are :

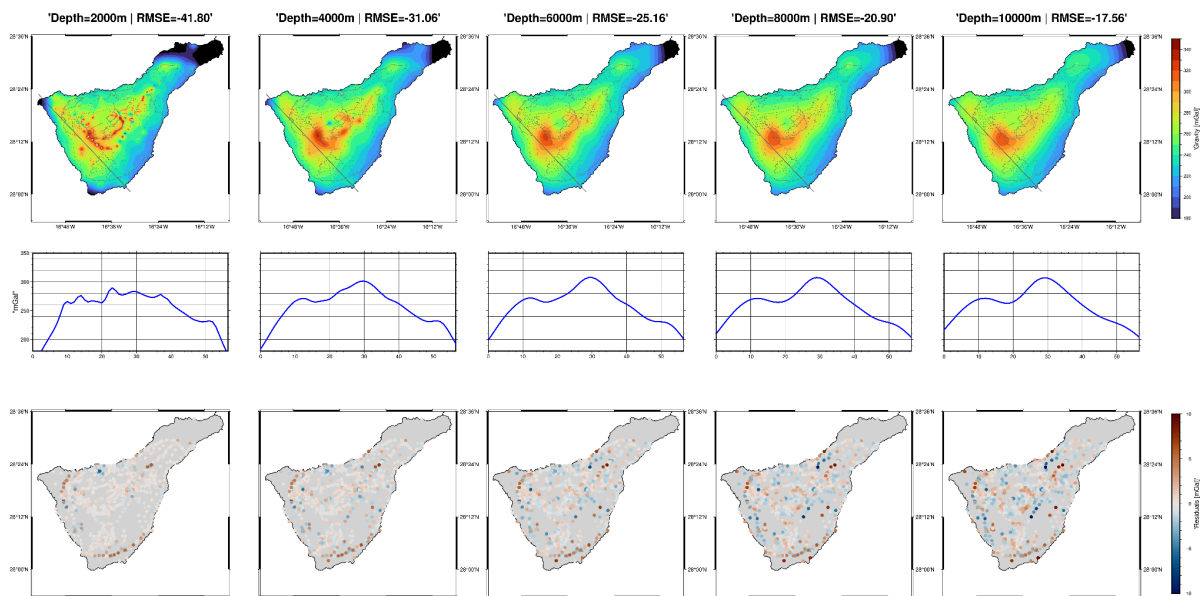
- **Damping** : It serves as a mathematical stabilizer during the inversion by penalizing extreme or abrupt mass variations between neighboring sources, which prevents the model from generating noisy oscillations or artifacts in areas where you lack station coverage. A low damping (0.01) has a best score but creates these artifacts, while big values (10-100) smooths the signal a lot and low score.
- **Depth** : It defines the vertical distance between the ground stations and the layer of equivalent sources, acting as a physical filter: shallow depths capture high-frequency, local geological features (like rifts), while greater depths damp out short wavelengths to map broader structures. But shallow (2000-4000) source depth creates artifacts (eyes) and is very sensitive to the datapoints, while deeper depth (6000-10000) smooths and increases the score.

Harmonica library has a script that estimate these ([Estimating damping and depth parameters | Harmonica v0.7.0](#)) but we didn't use it because it always selected extreme values (lowest damping & deepest depth) due to residual/score/RMSE calculation.

INFLUENCE OF DAMPING | Dataset: ALL | Fixed Depth: 6000m



INFLUENCE OF DEPTH | Dataset: ALL | Fixed Damping: 1



With these we can see the effect of depth and damping on the model and on the residuals. The increase of damping smoothen the isolines where we don't have data. A good value could be 0.1. or 1 (higher residual but smoother map). Shallow depth creates very sharp and localized signals under the data points while deeper creates a very smooth trend, but the deeper the more residuals (bc too smooth). A good value could be 6000 or 7000 m.

The real EQS interpolation is in one step with the residual separation in the next section : `FINAL_RESIDUAL_SEPARATION.ipynb`

The test of different hyperparameters and datasets helps to see if some of the features are mathematical artifacts or real features. It helps to select empirically the right parameters.

Final hyperparameters chosen: 7000m depth and 0.5 damping, project at ellipsoid height =0.